Run report: item4-run-20261018T2049Z

Descriptive record of one recorded run of examples/genus4_res_census.rs (Parts T, H4, K) under `research/weil_descent_angles_20261018/PROTOCOL.md` (Addendum A and Amendments B–G, each committed before the run it governs). It lists what ran, on what host, and what the tool recorded, file by file. Interpretation is in the study's `REPORT.md`. Costs are counted operations (F_p multiplications and inversions, group operations, native operations), never wall-clock; the only times are the tool's own seconds. Isolation: none earned (a shared cloud container, 4 threads); this affects only the seconds, since every recorded value is an exact count or a counted operation total.

env.txt

```
run: item 4, 20261018T2049Z (Parts T, H4, K)
protocol commit: bdcf8f8ad (item 4 frozen); tool commit: 252b52998
rustc 1.97.0 (2d8144b78 2026-07-07)
host: Linux 6.18.44-fc-v139 x86_64, 4 cores, Intel(R) Xeon(R) Processor @ 2.10GHz, 15 GB
command: rustc --edition 2021 -O examples/genus4_res_census.rs -o g4 && G4_THREADS=4 ./g4 research/weil_descent_angles_20261018/results/item4-run-20261018T2049Z T H4 K
tool sha256: 8c6597ea8d5eb73604f8f2d3d724df4275deefce75db8944c6dc2c2b6edcc27b
threads: 4
smoke run before this run (not evidence): G4_LIMIT=200000 (4 s), Parts T and H4 as here; the K slice showed the expected orbit-outside-slice K5 failures only.
```

tool_log.txt

```
partT: n with an odd-t solution and n >= 3: [] (P5 predicts none)
partT: n = 2 odd-t cycle types: [6] [3, 3]
partT: n = 4 even-t solutions all in powers of 2: true
field F_3^12 ready 0.2s
H4: squarefree degree-10 binary forms = 104976; mass = 104976/48 = 2187 (check K1 expects 2187: ok)
H4: Res-shaped models = 1100; t histogram {-18: 18, -10: 96, -8: 144, -6: 96, -2: 192, 0: 204, 2: 180, 4: 48, 8: 48, 10: 48, 16: 24, 18: 2}
H4: check K2 (t even and cycle types in powers of 2) violations = 0
classify split: 36 classes, K5 violations 0
part K split: forms 43046720 smooth 21830400 mass 9475.0000 res_forms 81216 classes 36 (416.4s)
classify nonsplit: 39 classes, K5 violations 0
part K nonsplit: forms 43046720 smooth 27299520 mass 9479.0000 res_forms 110880 classes 39 (714.3s)
classify cone: 83 classes, K5 violations 0
part K cone: forms 14348906 smooth 8503056 mass 6561.0000 res_forms 101520 classes 83 (267.0s)
K: non-hyperelliptic mass total = 25515.000000 (check K3 literature value q^9+q^8-q^6 = 25515: match)
K: Res-shaped classes = 158; t histogram (class counts) {-18: 3, -17: 2, -16: 3, -15: 1, -14: 8, -13: 6, -12: 2, -11: 7, -10: 3, -9: 1, -8: 14, -7: 3, -6: 11, -5: 9, -4: 5, -3: 15, -2: 11, -1:
2, 0: 4, 1: 4, 2: 17, 3: 2, 4: 9, 5: 2, 7: 6, 9: 3, 10: 4, 11: 1}
K: check K4 violations = 0; check K5 violations = 0
K: P6 prime-order classes (82 - t prime, elliptic trace exists) = 9 : ["split:t=11:#E=71:aut=1", "nonsplit:t=-1:#E=83:aut=1", "cone:t=-7:#E=89:aut=1", "cone:t=9:#E=73:aut=3",
"cone:t=9:#E=73:aut=1", "cone:t=9:#E=73:aut=1", "cone:t=-1:#E=83:aut=1", "cone:t=-7:#E=89:aut=1", "cone:t=-7:#E=89:aut=1"]
total 1398.1s
```

checks.txt

```
partT: n with an odd-t solution and n >= 3: [] (P5 predicts none)
partT: n = 2 odd-t cycle types: [6] [3, 3]
partT: n = 4 even-t solutions all in powers of 2: true
H4: squarefree degree-10 binary forms = 104976; mass = 104976/48 = 2187 (check K1 expects 2187: ok)
H4: Res-shaped models = 1100; t histogram {-18: 18, -10: 96, -8: 144, -6: 96, -2: 192, 0: 204, 2: 180, 4: 48, 8: 48, 10: 48, 16: 24, 18: 2}
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2, 0: 4, 1: 4, 2: 17, 3: 2, 4: 9, 5: 2, 7: 6, 9: 3, 10: 4, 11: 1}
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K: P6 prime-order classes (82 - t prime, elliptic trace exists) = 9 : ["split:t=11:#E=71:aut=1", "nonsplit:t=-1:#E=83:aut=1", "cone:t=-7:#E=89:aut=1", "cone:t=9:#E=73:aut=3",
"cone:t=9:#E=73:aut=1", "cone:t=9:#E=73:aut=1", "cone:t=-1:#E=83:aut=1", "cone:t=-7:#E=89:aut=1", "cone:t=-7:#E=89:aut=1"]
```

h4_res_shaped.csv (first 12 of 1100 rows)

code	a0..a10	N4	t	t_even	cycle_type	cycles_powers_of_2
19686	0 1 0 0 0 0 0 0 0 1 0	154	−18	true	1+1+4+4	true
19687	1 1 0 0 0 0 0 0 0 1 0	154	−18	true	1+1+4+4	true
19688	2 1 0 0 0 0 0 0 0 1 0	154	−18	true	1+1+4+4	true
19689	0 2 0 0 0 0 0 0 0 1 0	10	18	true	1+1+1+1+2+2+2	true

19716	0 2 0 1 0 0 0 0 0 1 0	82	0	true	1+1+8	true
19717	1 2 0 1 0 0 0 0 0 1 0	82	0	true	1+1+8	true
19718	2 2 0 1 0 0 0 0 0 1 0	82	0	true	1+1+8	true
19743	0 2 0 2 0 0 0 0 0 1 0	82	0	true	1+1+8	true
19744	1 2 0 2 0 0 0 0 0 1 0	82	0	true	1+1+8	true
19745	2 2 0 2 0 0 0 0 0 1 0	82	0	true	1+1+8	true
19932	0 2 0 0 0 1 0 0 0 1 0	18	16	true	1+1+8	true
19960	1 2 0 1 0 1 0 0 0 1 0	90	−2	true	1+1+8	true

k_classes.csv (first 12 of 158 rows)

type	rep_code	orbit_forms	aut	N4	t	numE	numE_prime	elliptic_trace_exists	form
split	708734	2304	1	94	−3	85	false	false	2*x^0y^0 + 1*x^0y^2 + 2*x^0y^3 + 1*x^1y^0 + 1*x^2y^3 + 1*x^3y^0
split	708761	2304	1	106	−6	88	false	false	2*x^0y^0 + 1*x^0y^2 + 2*x^1y^0 + 1*x^2y^3 + 1*x^3y^0
split	708989	2304	1	106	−6	88	false	false	2*x^0y^0 + 1*x^0y^1 + 2*x^0y^2 + 2*x^0y^3 + 1*x^1y^0 + 1*x^1y^1 + 1*x^2y^3 + 1*x^3y^0
split	728311	2304	1	114	−8	90	false	true	1*x^0y^0 + 1*x^0y^1 + 1*x^0y^2 + 1*x^0y^3 + 1*x^2y^1 + 1*x^2y^3 + 1*x^3y^0
split	729470	2304	1	90	−2	84	false	true	2*x^0y^0 + 1*x^0y^2 + 2*x^0y^3 + 2*x^1y^0 + 1*x^1y^1 + 1*x^1y^2 + 1*x^2y^1 + 1*x^2y^3 + 1*x^3y^0
split	729949	2304	1	102	−5	87	false	true	1*x^0y^0 + 1*x^0y^1 + 2*x^0y^3 + 2*x^1y^0 + 2*x^1y^2 + 1*x^2y^1 + 1*x^2y^3 + 1*x^3y^0
split	755479	2304	1	54	7	75	false	true	1*x^0y^0 + 2*x^0y^2 + 2*x^0y^3 + 2*x^1y^0 + 1*x^1y^2 + 1*x^2y^0 + 2*x^2y^1 + 1*x^2y^3 + 1*x^3y^0
split	768113	2304	1	78	1	81	false	true	2*x^0y^0 + 2*x^0y^1 + 1*x^0y^2 + 2*x^0y^3 + 2*x^1y^0 + 1*x^1y^1 + 1*x^2y^2 + 1*x^2y^3 + 1*x^3y^0
split	768506	2304	1	94	−3	85	false	false	2*x^0y^0 + 1*x^0y^1 + 2*x^0y^3 + 1*x^1y^0 + 1*x^1y^2 + 1*x^2y^2 + 1*x^2y^3 + 1*x^3y^0
split	769274	2304	1	66	4	78	false	true	2*x^0y^0 + 2*x^0y^1 + 1*x^0y^2 + 2*x^1y^0 + 2*x^1y^2 + 1*x^2y^2 + 1*x^2y^3 + 1*x^3y^0
split	769336	2304	1	94	−3	85	false	false	1*x^0y^0 + 2*x^0y^1 + 2*x^0y^2 + 2*x^0y^3 + 2*x^1y^0 + 2*x^1y^2 + 1*x^2y^2 + 1*x^2y^3 + 1*x^3y^0
split	774323	2304	1	38	11	71	true	true	2*x^0y^0 + 2*x^0y^1 + 1*x^0y^2 + 1*x^0y^3 + 1*x^1y^0 + 1*x^2y^0 + 1*x^2y^2 + 1*x^2y^3 + 1*x^3y^0

k_summary.csv (first 3 of 3 rows)

type	forms	smooth	mass_den	mass	smooth_with_N1_4	res_shaped_forms	res_shaped_classes	k5_violations	seconds
split	43046720	21830400	2304	9475.000000	4100256	81216	36	0	416.4
nonsplit	43046720	27299520	2880	9479.000000	5611680	110880	39	0	714.3
cone	14348906	8503056	1296	6561.000000	1925208	101520	83	0	267.0

part_t.csv (first 12 of 30 rows)

n	t_parity	solutions	cycle_types
2	odd	2	6 3+3
2	even	6	4+2 4+1+1 2+2+2 2+2+1+1 2+1+1+1+1 1+1+1+1+1+1
3	odd	0	
3	even	4	6+2 6+1+1 3+3+2 3+3+1+1
4	odd	0	
4	even	14	8+2 8+1+1 4+4+2 4+4+1+1 4+2+2+2 4+2+2+1+1 4+2+1+1+1+1 4+1+1+1+1+1+1 2+2+2+2+2 2+2+2+2+1+1 2+2+2+1+1+1+1 2+2+1+1+1+1+1+1 2+1+1+1+1+1+1+1+1 1+1+1+1+1+1+1+1+1+1
5	odd	0	

5	even	4	10+2 10+1+1 5+5+2 5+5+1+1
6	odd	0	
6	even	8	12+2 12+1+1 6+6+2 6+6+1+1 6+3+3+2 6+3+3+1+1 3+3+3+3+2 3+3+3+3+1+1
7	odd	0	
7	even	4	14+2 14+1+1 7+7+2 7+7+1+1

Graph

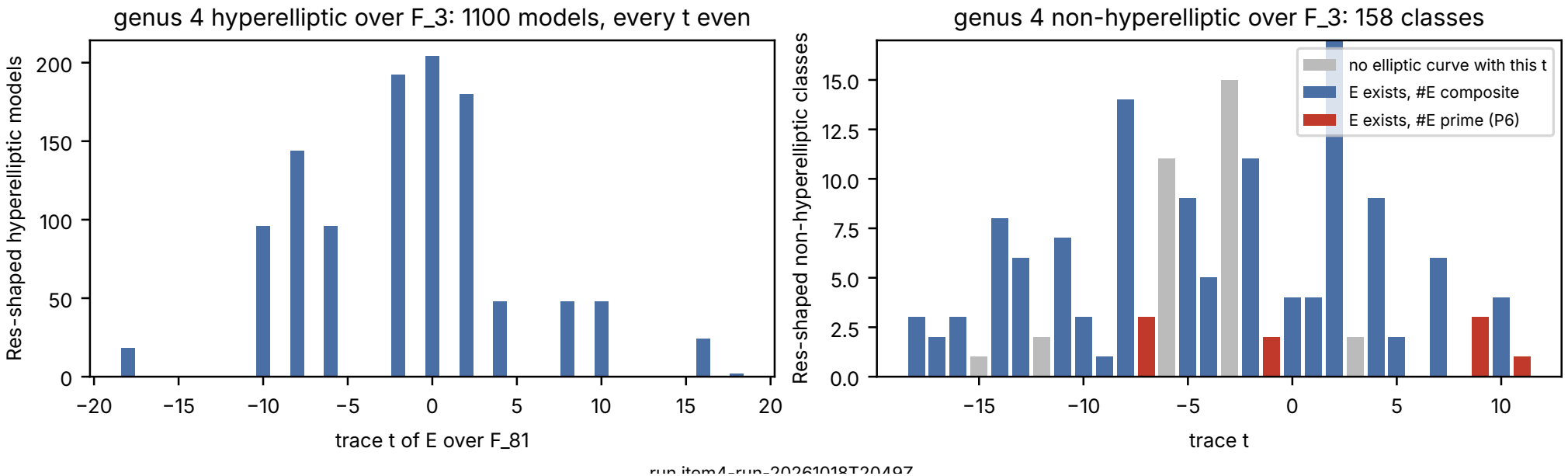


Figure 1: Res-shaped genus-4 curves over F_3 by trace: hyperelliptic models (left), non-hyperelliptic classes (right); exact census.